COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 2 : Session – 1 : SLO – 1 | Functional units of a computer
Name: JATANPREET SINGH VIRDI JATANPREET SINGH VIRDI | Reg. No: RA2511003011374 | Solved Worksheet

Section A – Match the Following (Answers)

- 1. Input Unit → a. Accepts data and instructions
- 2. Output Unit → b. Displays processed data
- 3. ALU → c. Performs arithmetic and logic operations
- 4. Control Unit → d. Controls and coordinates activities
- 5. Memory Unit → e. Stores data and instructions

Section B – Fill in the Blanks

- The INPUT unit accepts data from the user.
- The ALU performs mathematical calculations and logical decisions.
- MEMORY stores data, instructions, and results.
- The CONTROL UNIT manages and coordinates all other units.
- The OUTPUT unit provides output information to the user.

Section C – Activity Task

- Block diagram (described): Input Unit → Memory Unit ■ CPU (Control Unit + ALU) → Output Unit, with the Control Unit connected to and coordinating every block.
- Data flow: data and instructions enter through the Input Unit and are stored in Memory; the Control Unit fetches instructions from memory, decodes them and directs the ALU to process the data; intermediate and final results return to Memory; the Output Unit presents the results to the user.
- CU and ALU working together: The Control Unit fetches and decodes each instruction, then sends control signals telling the ALU which arithmetic/logic operation to perform and on which data. The ALU executes the operation and flags the result status back to the Control Unit, which then decides the next step. Thus the CU orchestrates while the ALU computes — together they execute instructions.

Section D – True or False

- The ALU is responsible for storing data permanently. — FALSE
- The Control Unit directs the operation of the processor. — TRUE
- Input unit converts results from machine form to human readable form. — FALSE
- The memory unit only stores the final output. — FALSE
- Output unit displays or prints the results of processing. — TRUE